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Power conversion system with increased detection accuracy

Abstract

A power conversion apparatus includes an input power terminal, an output power terminal, a power conversion circuit, a signal generation circuit, and a control circuit. The power conversion circuit converts supplied electric power and outputs the converted electric power. The power conversion circuit includes a sensor that generates a first detection signal corresponding to an output voltage or an output current. The signal generation circuit generates a second detection signal corresponding to the first detection signal. The control circuit controls operation of the power conversion circuit. The signal generation circuit includes a corrector, a photocoupler, and an output circuit. The corrector performs correction processing on the first detection signal. The photocoupler includes a light receiving device generating a light reception signal. The output circuit outputs the second detection signal corresponding to the light reception signal. The corrector performs the correction processing corresponding to a current transfer ratio of the photocoupler.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) The present application claims priority from Japanese Patent Application No. 2022-084537 filed on May 24, 2022, the entire contents of which are hereby incorporated by reference.
BACKGROUND
(2) The disclosure relates to a power conversion apparatus and a power conversion system that each convert electric power.
(3) An exemplary power conversion apparatus detects an output voltage or an output current. For example, Japanese Unexamined Patent Application Publication No. 2019-092244 discloses a technique of controlling respective output currents of parallel-coupled power conversion apparatuses to be substantially equal to each other.
SUMMARY
(4) A power conversion apparatus according to an embodiment of the disclosure includes an input power terminal, an output power terminal, a power conversion circuit, a signal generation circuit, and a control circuit. The power conversion circuit is configured to convert electric power supplied via the input power terminal and to output converted electric power via the output power terminal.

The power conversion circuit includes a sensor configured to generate a first detection signal corresponding to an output voltage or an output current. The signal generation circuit is configured to generate a second detection signal corresponding to the first detection signal. The control circuit is configured control operation of the power conversion circuit. The signal generation circuit includes a corrector, a photocoupler, and an output circuit. The corrector is configured to perform correction processing on the first detection signal. The photocoupler includes a light emitting device and a light receiving device. The light emitting device is configured to emit light at a luminance corresponding to the first detection signal having undergone the correction processing. The light receiving device is configured to receive the light emitted by the light emitting device and to generate a light reception signal corresponding to the amount of the light received from the light emitting device. The output circuit is configured to output the second detection signal corresponding to the light reception signal. The corrector is configured to perform the correction processing corresponding to a current transfer ratio of the photocoupler.
(5) A power conversion system according to an embodiment of the disclosure includes a plurality of power conversion apparatuses. Each of the power conversion apparatuses includes an input power terminal, an output power terminal, a power conversion circuit, a signal generation circuit, a signal terminal, and a control circuit. The power conversion circuit is configured to convert electric power supplied via the input power terminal and to output converted electric power via the output power terminal. The power conversion circuit includes a sensor configured to generate a first detection signal corresponding to an output voltage or an output current. The signal generation circuit is configured to generate a second detection signal corresponding to the first detection signal. The signal terminal is led to a terminal that allows for output of the second detection signal of the signal generation circuit. The control circuit is configured control operation of the power conversion circuit, based on a voltage at the signal terminal and a voltage of the second detection signal. The respective signal terminals of the power conversion apparatuses are coupled to each other. The signal generation circuit includes a corrector, a photocoupler, and an output circuit. The corrector is configured to perform correction processing on the first detection signal. The photocoupler includes a light emitting device and a light receiving device. The light emitting device is configured to emit light at a luminance corresponding to the first detection signal having undergone the correction processing. The light receiving device is configured to receive the light emitted by the light emitting device and to generate a light reception signal corresponding to the amount of the light received from the light emitting device. The output circuit is configured to output the second detection signal corresponding to the light reception signal. The corrector is configured to perform the correction processing corresponding to a current transfer ratio of the photocoupler.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments and, together with the specification, serve to explain the principles of the disclosure.
(2) FIG. **1** is a block diagram illustrating a configuration example of a power conversion system according to one example embodiment of the disclosure.
(3) FIG. **2** is an explanatory diagram illustrating a coupling example of power conversion apparatuses illustrated in FIG. **1**.
(4) FIG. **3** is a block diagram illustrating a configuration example of each of the power conversion apparatuses illustrated in FIG. **1**.
(5) FIG. **4** is a circuit diagram illustrating a configuration example of an isolated power conversion

circuit illustrated in FIG. **3**.
(6) FIG. **5** is a circuit diagram illustrating a configuration example of a corrector and an output circuit illustrated in FIG. **3**.
(7) FIG. **6** is a circuit diagram illustrating a configuration example of another corrector and another output circuit illustrated in FIG. **3**.
(8) FIG. **7** is a waveform diagram illustrating a characteristic example of a signal generation circuit illustrated in FIG. **5**.
(9) FIG. **8** is a waveform diagram illustrating a characteristic example of a signal generation circuit according to a comparative example.
(10) FIG. **9** is a block diagram illustrating a configuration example of a power conversion system according to a modification example.
(11) FIG. **10** is a block diagram illustrating a configuration example of each of power conversion apparatuses illustrated in FIG. **9**.
(12) FIG. **11** is a circuit diagram illustrating a configuration example of a non-isolated power conversion circuit illustrated in FIG. **10**.
(13) FIG. **12** is a block diagram illustrating a configuration example of a power conversion system according to another modification example.
(14) FIG. **13** is an explanatory diagram illustrating a coupling example of power conversion apparatuses illustrated in FIG. **12**.

DETAILED DESCRIPTION

(15) It is desired that an output voltage and an output current be detected with high accuracy, and expectations are placed on a further improvement in detection accuracy.
(16) It is desirable to provide a power conversion apparatus and a power conversion system that each make it possible to increase detection accuracy.
(17) In the following, some example embodiments of the disclosure are described in detail with reference to the accompanying drawings. Note that the following description is directed to illustrative examples of the disclosure and not to be construed as limiting to the disclosure. Factors including, without limitation, numerical values, shapes, materials, components, positions of the components, and how the components are coupled to each other are illustrative only and not to be construed as limiting to the disclosure. Further, elements in the following example embodiments which are not recited in a most-generic independent claim of the disclosure are optional and may be provided on an as-needed basis. The drawings are schematic and are not intended to be drawn to scale. Throughout the present specification and the drawings, elements having substantially the same function and configuration are denoted with the same reference numerals to avoid any redundant description. In addition, elements that are not directly related to any embodiment of the disclosure are unillustrated in the drawings.

EXAMPLE EMBODIMENT

Configuration Example

(18) FIG. **1** illustrates a configuration example of a power conversion system **1** according to an example embodiment of the disclosure. The power conversion system **1** may be configured to convert a direct-current electric power supplied from a direct-current power supply PDC and to supply the converted direct-current electric power to a load LD.
(19) The power conversion system **1** may include power terminals T**11** and T**12**, a plurality of power conversion apparatuses **10**, and power terminals T**21** and T**22**. In the example embodiment, the power conversion system **1** may include four power conversion apparatuses **10**A, **10**B, **10**C, and **10**D, although it is not limited thereto.
(20) The power terminals T**11** and T**12** may each allow for reception of electric power to the power conversion system **1**. The power terminal T**11** may be coupled to a first end of the direct-current power supply PDC, and the power terminal T**12** may be coupled to a second end of the direct-current power supply PDC. For example, the direct-current power supply PDC may be a power

supply circuit that generates direct-current electric power, or may be a battery.
(21) The four power conversion apparatuses **10** may each be an isolated DC-to-DC conversion circuit, and may each include input power terminals Vip and Vin, output power terminals Vop and Von, balance terminals Vbi and Vbv, and a reference terminal GNDb.
(22) The input power terminals Vip and Vin may each allow for reception of electric power to the power conversion apparatus **10**. The respective input power terminals Vip of the power conversion apparatuses **10**A to **10**D may be coupled to each other, and may also be coupled to the power terminal T**11**. The respective input power terminals Vin of the power conversion apparatuses **10**A to **10**D may be coupled to each other, and may also be coupled to the power terminal T**12**.
(23) The output power terminals Vop and Von may each allow for output of electric power converted by the power conversion apparatus **10**. The respective output power terminals Vop of the power conversion apparatuses **10**A and **10**B may be coupled to each other, and may also be coupled to the power terminal T**21**. The respective output power terminals Von of the power conversion apparatuses **10**A and **10**B may be coupled to each other, and may also be coupled to the respective output power terminals Vop of the power conversion apparatuses **10**C and **10**D. The respective output power terminals Vop of the power conversion apparatuses **10**C and **10**D may be coupled to each other, and may also be coupled to the respective output power terminals Von of the power conversion apparatuses **10**A and **10**B. The respective output power terminals Von of the power conversion apparatuses **10**C and **10**D may be coupled to each other, and may also be coupled to the power terminal T**22**.
(24) The balance terminal Vbi may allow for reception and output of a signal adapted to cause respective output currents, i.e., later-described output currents Tout, of the four power conversion apparatuses **10** to be equal to each other. The respective balance terminals Vbi of the power conversion apparatuses **10**A to **10**D are couped to each other.
(25) The balance terminal Vbv may allow for reception and output of a signal adapted to cause respective output voltages, i.e., later-described output voltages Vout, of the four power conversion apparatuses **10** to be equal to each other. The respective balance terminals Vbv of the power conversion apparatuses **10**A to **10**D are couped to each other.
(26) The reference terminal GNDb may allow for reception and output of a reference voltage of later-described tertiary-side circuitry of corresponding one of the four power conversion apparatuses **10**. The respective reference terminals GNDb of the power conversion apparatuses **10**A to **10**D may be couped to each other.
(27) The power terminals T**21** and T**22** may each allow for output of electric power converted by the power conversion system **1**. The power terminal T**21** may be coupled to a first end of the load LD, and the power terminal T**22** may be coupled to a second end of the load LD.
(28) FIG. **2** illustrates coupling between the output power terminals Vop and Von of the four power conversion apparatuses **10**. The respective output power terminals Vop of the power conversion apparatuses **10**A and **10**B may be coupled to a power node N**1** led to the power terminal T**21**. The respective output power terminals Von of the power conversion apparatuses **10**A and **10**B may be coupled to a power node N**2**. The respective output power terminals Vop of the power conversion apparatuses **10**C and **10**D may be coupled to the power node N**2**. The respective output power terminals Von of the power conversion apparatuses **10**C and **10**D may be coupled to a power node N**3** led to the power terminal T**22**. In the power conversion system **1**, the power conversion apparatuses **10**A and **10**B may be coupled in parallel, and the power conversion apparatuses **10**C and **10**D may be coupled in parallel. Further, the power conversion apparatuses **10**A and **10**B and the power conversion apparatuses **10**C and **10**D may be coupled in series.
(29) [Power Conversion Apparatus **10**]
(30) FIG. **3** illustrates a configuration example of the power conversion apparatus **10**. The power conversion apparatus **10** may include an isolated power conversion circuit **11**, correction circuits **12**A and **12**B, photocouplers **13**A and **13**B, correction circuits **14**A and **14**B, resistors **15**A and **15**B,

amplifiers **16**A and **16**B, a microcontroller unit (an MCU) **30**, a digital isolator **17**, an MCU **20**, and an isolated drive circuit **18**.
(31) The isolated power conversion circuit **11** is configured to convert direct-current electric power supplied via the input power terminals Vip and Vin and to output the converted direct-current electric power via the output power terminals Vop and Von.
(32) FIG. **4** illustrates a configuration example of the isolated power conversion circuit **11**. The isolated power conversion circuit **11** may include a capacitor C**1**, transistors SW**1** and SW**2**, a transformer TR, a rectifier circuit **43**, a current sensor **44**, a smoothing circuit **45**, and a voltage sensor **46**. Primary-side circuitry of the isolated power conversion circuit **11** may include the capacitor C**1** and the transistors SW**1** and SW**2**. Secondary-side circuitry of the isolated power conversion circuit **11** may include the rectifier circuit **43**, the current sensor **44**, the smoothing circuit **45**, and the voltage sensor **46**.
(33) The capacitor C**1** may have a first coupled to a voltage line L**11** led to the input power terminal Vip, and a second end coupled to a reference voltage line L**12** led to the input power terminal Vin. In the example embodiment, the transistors SW**1** and SW**2** may be N-type field-effect transistors (FETs), although they are not limited thereto. The transistor SW**1** may have a gate to be supplied with a gate signal SG, a drain coupled to a later-described primary winding **41** of the transformer TR, and a source coupled to the reference voltage line L**12**. The transistor SW**2** may have a gate to be supplied with the gate signal SG, a drain coupled to the primary winding **41** of the transformer TR, and a source coupled to the reference voltage line L**12**.
(34) The transformer TR may be configured to provide direct-current isolation and alternating-current coupling between the primary-side circuitry and the secondary-side circuitry. The transformer TR may include the primary winding **41** and a secondary winding **42**. The primary winding **41** may have a first end coupled to the voltage line L**11**, and a second end coupled to the drains of the transistors SW**1** and SW**2**. The secondary winding **42** may have a first end coupled to a voltage line L**21** led to the output power terminal Vop, and a second end coupled to a reference voltage line L**22** led to the output power terminal Von.
(35) The rectifier circuit **43** may be configured to rectify an alternating-current voltage outputted from the secondary winding **42** of the transformer TR. The rectifier circuit **43** may include diodes D**1** and D**2**. The diode D**1** may be provided on the voltage line L**21**. The diode D**1** may have an anode coupled to the first end of the secondary winding **42**, and a cathode coupled to a cathode of the diode D**2** and to the smoothing circuit **45**. The diode D**2** may have an anode coupled to the reference voltage line L**22**, and the cathode coupled to the cathode of the diode D**1** on the reference voltage line L**21**.
(36) The current sensor **44** may be configured to detect the output current Tout of the power conversion apparatus **10**. The current sensor **44** may be provided on the reference voltage line L**22**. The current sensor **44** may have a first end coupled to the second end of the secondary winding **42** and to the anode of the diode D**2**, and a second end coupled to the smoothing circuit **45**. On the reference voltage line L**22**, the current sensor **44** may detect a current flowing from the smoothing circuit **45** toward the rectifier circuit **43** as the output current Tout. The output current Tout may have a positive polarity when the current is directed from the smoothing circuit **45** toward the rectifier circuit **43**. In addition, the current sensor **44** may be configured to generate a detection signal SI having a voltage corresponding to the output current Tout. The current sensor **44** may include, for example, a resistor provided on the reference voltage line L**22** and an amplifier circuit amplifying a voltage difference across the resistor. In such a case, the current sensor **44** may generate the detection signal SI by amplifying the voltage difference across the resistor. Note that although the current sensor **44** may be provided on the reference voltage line L**22** in the example embodiment, this is non-limiting. In some embodiments, the current sensor **44** may be provided on the voltage line L**21**.
(37) The smoothing circuit **45** may be configured to smooth the voltage rectified by the rectifier

circuit **43**. The smoothing circuit **45** may include an inductor L**1** and a capacitor C**2**. The inductor L**1** may be provided on the voltage line L**21**. The inductor L**1** may have a first end coupled to the cathodes of the diodes D**1** and D**2**, and a second end coupled to the capacitor C**2**. The capacitor C**2** may have a first end coupled to the second end of the inductor L**1** on the voltage line L**21**, and a second end coupled to the second end of the current sensor **44** on the reference voltage line L**22**.
(38) The voltage sensor **46** may be configured to detect the output voltage Vout of the power conversion apparatus **10**. The voltage sensor **46** may have a first end coupled to the second end of the inductor L**1** on the voltage line L**21**, and a second end coupled to the second end of the current sensor **44** on the reference voltage line L**22**. The voltage sensor **46** may detect, as the output voltage Vout, a voltage, at the voltage line L**21**, which is based on a voltage at the reference voltage line L**22**. In addition, the voltage sensor **46** may be configured to generate a detection signal SV having a voltage value corresponding to a voltage value of the above-described voltage. The voltage sensor **46** may include, for example, a resistor network including multiple resistors coupled in series to each other. In such a case, the voltage sensor **46** may generate the detection signal SV by dividing the output voltage Vout using the multiple resistors.
(39) As illustrated in FIG. **1**, the four power conversion apparatuses **10**A to **10**D may be supplied with direct-current electric power from the direct-current power supply PDC. Accordingly, respective operating voltages of primary-side circuitry of the power conversion apparatuses **10**A to **10**D may be equal to each other.
(40) As illustrated in FIG. **2**, the power conversion apparatuses **10**A and **10**B and the power conversion apparatuses **10**C and **10**D may be coupled in series. Accordingly, respective operating voltages, based on a voltage at the power terminal T**22**, of secondary-side circuitry of the power conversion apparatuses **10**A and **10**B may be higher than respective operating voltages, based on the voltage at the power terminal T**22**, of the secondary-side circuitry of the power conversion apparatuses **10**C and **10**D.
(41) In FIG. **3**, the correction circuits **12**A and **12**B and the MCU **20** may configure the secondary-side circuitry, together with the rectifier circuit **43**, the current sensor **44**, the smoothing circuit **45**, and the voltage sensor **46** illustrated in FIG. **4**. The correction circuits **14**A and **14**B, the resistors **15**A and **15**B, the amplifiers **16**A and **16**B, and the MCU **30** may configure the tertiary-side circuitry.
(42) As illustrated in FIG. **1**, the reference terminals GNDb of the four power conversion apparatuses **10** may be coupled to each other. Accordingly, reference voltages of the tertiary-side circuitry of the four power conversion apparatuses **10** may be the same, and respective operating voltages of the tertiary-side circuitry of the four power conversion apparatuses **10** may be equal to each other accordingly.
(43) As illustrated in FIG. **3**, the MCU **20** may include correction arithmetic circuits **21**A and **21**B, error amplifiers **22**A and **22**B, and a switching control circuit **23**. The MCU **30** may include correction arithmetic circuits **31**A and **31**B, adjustment arithmetic circuits **32**A and **32**B, a command value generation circuit **33**, and adder circuits **34**A and **34**B. The MCUs **20** and **30** may each include an analog-to-digital conversion circuit that converts a supplied analog signal into a digital value. The MCUs **20** and **20** may each perform processing, based on the digital value resulting from the conversion.
(44) The correction circuit **12**A, and the correction arithmetic circuit **21**A of the MCU **20** may configure a corrector **101**A.
(45) The correction circuit **12**A may be configured to correct the detection signal SI supplied from the current sensor **44**. The correction circuit **12**A may subject the detection signal SI to correction corresponding to a characteristic of the photocoupler **13**A, based on a correction signal SCI supplied from the correction arithmetic circuit **21**A. Based on the detection signal SI having undergone the correction, the correction circuit **12**A may drive a light emitting device of the photocoupler **13**A.

(46) The correction arithmetic circuit **21**A may be configured to recognize the characteristic of the photocoupler **13**A by transmitting and receiving data to and from the correction arithmetic circuit **31**A via the digital isolator **17**, and to generate the correction signal SCI corresponding to the characteristic of the photocoupler **13**A. Further, the correction arithmetic circuit **21**A may supply the correction circuit **12**A with the correction signal SCI generated.
(47) The photocoupler **13**A may be configured to transmit and receive signals while electrically isolating the signals. The photocoupler **13**A includes the light emitting device and a light receiving device. The light emitting device may be coupled to the correction circuit **12**A, and the light receiving device may be coupled to the correction circuit **14**A. The light emitting device emits light at a luminance corresponding to a signal supplied from the correction circuit **12**A. The light receiving device receives the light emitted by the light emitting device. The light receiving device may supply the correction circuit **14**A with a light reception signal corresponding to the amount of the light received from the light emitting device.
(48) The correction circuit **14**A, and the correction arithmetic circuit **31**A of the MCU **30** may configure an output circuit **102**A.
(49) The correction circuit **14**A may be configured to generate a detection signal SI**2** corresponding to the light reception signal supplied from the photocoupler **13**A, and to correct the detection signal SI**2**, based on a correction signal SCI**2** supplied from the correction arithmetic circuit **31**A. The correction circuit **14**A may perform correction corresponding to the characteristic of the photocoupler **13**A.
(50) The correction arithmetic circuit **31**A may be configured to estimate the characteristic of the photocoupler **13**A by transmitting and receiving data to and from the correction arithmetic circuit **21**A via the digital isolator **17**, and to generate the correction signal SCI**2** corresponding to the characteristic of the photocoupler **13**A. Further, the correction arithmetic circuit **31**A may supply the correction circuit **14**A with the correction signal SCI**2** generated.
(51) Such a configuration makes it possible for the power conversion apparatus **10** to generate, based on the detection signal SI supplied from the current sensor **44**, the detection signal SI**2** corresponding to the detection signal SI, while reducing an influence of the characteristic of the photocoupler **13**A. For example, the photocoupler **13**A can vary in current transfer ratio (CTR) depending on an environmental condition, such as temperature. The current transfer ratio is a ratio of a light reception current flowing through the light receiving device of the photocoupler **13**A to a light emission current flowing through the light emitting device of the photocoupler **13**A. The current transfer ratio can also decrease, for example, with time. Because the current transfer ratio of the photocoupler **13**A can thus vary, the detection signal SI**2** can vary even if the detection signal SI is the same. In the power conversion apparatus **10**, the detection signal SI**2** may be generated by performing correction corresponding to the characteristic of the photocoupler **13**A. This makes it possible for the power conversion apparatus **10** to reduce the influence of the characteristic of the photocoupler **13**A on the detection signal SI**2**.
(52) The correction circuit **12**B, and the correction arithmetic circuit **21**B of the MCU **20** may configure a corrector **101**B.
(53) As with the correction circuit **12**A, the correction circuit **12**B may be configured to correct the detection signal SV supplied from the voltage sensor **46**. The correction circuit **12**B may subject the detection signal SV to correction corresponding to a characteristic of the photocoupler **13**B, based on a correction signal SCV supplied from the correction arithmetic circuit **21**B. Based on the detection signal SV having undergone the correction, the correction circuit **12**B may drive a light emitting device of the photocoupler **13**B.
(54) As with the correction arithmetic circuit **21**A, the correction arithmetic circuit **21**B may be configured to recognize the characteristic of the photocoupler **13**B by transmitting and receiving data to and from the correction arithmetic circuit **31**B via the digital isolator **17**, and to generate the correction signal SCV corresponding to the characteristic of the photocoupler **13**B. Further, the

correction arithmetic circuit **21**B may supply the correction circuit **12**B with the correction signal SCV generated.

(55) As with the photocoupler **13**A, the photocoupler **13**B may be configured to transmit and receive signals while electrically isolating the signals. The photocoupler **13**B includes the light emitting device and a light receiving device. The light emitting device may be coupled to the correction circuit **12**B, and the light receiving device may be coupled to the correction circuit **14**B.

(56) The correction circuit **14**B, and the correction arithmetic circuit **31**B of the MCU **30** may configure an output circuit **102**B.

(57) As with the correction circuit **14**A, the correction circuit **14**B may be configured to generate a detection signal SV**2** corresponding to a light reception signal supplied from the photocoupler **13**B, and to correct the detection signal SV**2**, based on a correction signal SCV**2** supplied from the correction arithmetic circuit **31**B. The correction circuit **14**B may perform correction corresponding to the characteristic of the photocoupler **13**B.

(58) As with the correction arithmetic circuit **31**A, the correction arithmetic circuit **31**B may be configured to estimate the characteristic of the photocoupler **13**B by transmitting and receiving data to and from the correction arithmetic circuit **21**B via the digital isolator **17**, and to generate the correction signal SCV**2** corresponding to the characteristic of the photocoupler **13**B. Further, the correction arithmetic circuit **31**B may supply the correction circuit **14**B with the correction signal SCV**2** generated.

(59) Such a configuration makes it possible for the power conversion apparatus **10** to generate, based on the detection signal SV supplied from the voltage sensor **46**, as with the detection signal SI, the detection signal SV**2** corresponding to the detection signal SV, while reducing an influence of the characteristic of the photocoupler **13**B.

(60) The resistor **15**A may have a first end coupled to an output terminal of the correction circuit **14**A and to a negative input terminal of the amplifier **16**A, and a second end coupled to a positive input terminal of the amplifier **16**A and to the balance terminal Vbi. As illustrated in FIG. **1**, the respective balance terminals Vbi of the four power conversion apparatuses **10** are coupled to each other. This causes a voltage at the balance terminal Vbi to be an average voltage of the detection signals SI**2** of the four power conversion apparatuses **10**.

(61) The positive input terminal of the amplifier **16**A may be coupled to the second end of the resistor **15**A and to the balance terminal Vbi. The negative input terminal of the amplifier **16**A may be coupled to the first end of the resistor **15**A and to the output terminal of the correction circuit **14**A. The amplifier **16**A may amplify a voltage difference across the resistor **15**A to thereby generate a difference signal Sdfi.

(62) The resistor **15**B may have a first end coupled to an output terminal of the correction circuit **14**B and to a negative input terminal of the amplifier **16**B, and a second end coupled to a positive input terminal of the amplifier **16**B and to the balance terminal Vbv. As illustrated in FIG. **1**, the respective balance terminals Vbv of the four power conversion apparatuses **10** are coupled to each other. This causes a voltage at the balance terminal Vbv to be an average voltage of the detection signals SV**2** of the four power conversion apparatuses **10**.

(63) The positive input terminal of the amplifier **16**B may be coupled to the second end of the resistor **15**B and to the valance terminal Vbv. The negative input terminal of the amplifier **16**B may be coupled to the first end of the resistor **15**B and to the output terminal of the correction circuit **14**B. The amplifier **16**B may amplify a voltage difference across the resistor **15**B to thereby generate a difference signal Sdfv.

(64) The adjustment arithmetic circuit **32**A may be configured to generate a difference value Cdfi by performing predetermined adjustment arithmetic processing, based on a digital value of the difference signal Sdfi. The difference value Cdfi may vary depending on the difference signal Sdfi. The adjustment arithmetic circuit **32**B may be configured to generate a difference value Cdfv by performing predetermined adjustment arithmetic processing, based on a digital value of the

difference signal Sdfv. The difference value Cdfv may vary depending on the difference signal Sdfv.
(65) The command value generation circuit **33** may be configured to generate a voltage command value CMV of the output voltage Vout of the power conversion apparatus **10** and a current command value CMI of the output current Tout of the power conversion apparatus **10**.
(66) The adder circuit **34**A may be configured to generate a voltage command value CMV**2** by adding up the difference value Cdfi and the voltage command value CMV. The adder circuit **34**B may be configured to generate a current command value CMI**2** by adding up the difference value Cdfv and the current command value CMI.
(67) The digital isolator **17** may be configured to supply a digital signal from the MCU **20** to the MCU **30** and to supply a digital signal from the MCU **30** to the MCU **20**. The digital isolator **17** may transmit and receive the digital signals to and from the MCU **20** and the MCU **30** while electrically isolating the MCU **20** and the MCU **30** from each other.
(68) The error amplifier **22**A may have a positive input terminal and a negative input terminal. The positive input terminal may receive the voltage command value CMV**2** supplied from the MCU **30** via the digital isolator **17**. The negative input terminal may receive a digital value of the detection signal SV supplied from the voltage sensor **46**. The digital value of the detection signal SV may correspond to a digital value of the output voltage Vout of the power conversion apparatus **10**. The error amplifier **22**A may generate an error value Cerv by amplifying a difference between the voltage command value CMV**2** and the digital value of the detection signal SV.
(69) The error amplifier **22**B may have a positive input terminal and a negative input terminal. The positive input terminal may receive the current command value CMI**2** supplied from the MCU **30** via the digital isolator **17**. The negative input terminal may receive a digital value of the detection signal SI supplied from the current sensor **44**. The digital value of the detection signal SI may correspond to a digital value of the output current Iout of the power conversion apparatus **10**. The error amplifier **22**B may generate an error value Ceri by amplifying a difference between the current command value CMI**2** and the digital value of the detection signal SI.
(70) The switching control circuit **23** may be configured to generate a gate signal SG**1**, based on the error values Cerv and Ceri, and to control, based on the gate signal SG**1**, operation of the isolated power conversion circuit **11**.
(71) The isolated drive circuit **18** may be configured to generate the gate signal SG, based on the gate signal SG**1**, and to drive, based on the gate signal SG, the transistors SW**1** and SW**2** of the isolated power conversion circuit **11** illustrated in FIG. **4**. The MCU **20** generating the gate signal SG**1** may be an element of the secondary-side circuitry, and the transistors SW**1** and SW**2** to be supplied with the gate signal SG may be elements of the primary-side circuitry. The isolated drive circuit **18** may thus drive the transistors SW**1** and SW**2**, based on the gate signal SG**1**, while electrically isolating the MCU **20** and the transistors SW**1** and SW**2** from each other.
(72) With such a configuration, in the power conversion apparatus **10**, negative feedback control may be performed to cause the output voltage Vout detected by the voltage sensor **46** illustrated in FIG. **4** to be equal to a voltage indicated by the voltage command value CMV**2**, and negative feedback control may be performed to cause the output current Iout detected by the current sensor **44** to be equal to a current indicated by the current command value CMI**2**. The negative feedback control may be proportional (P) control or proportional and integral (PI) control.
(73) In the power conversion system **1**, it is possible to maintain balance between the respective output currents Iout of the power conversion apparatuses **10**A to **10**D. For example, suppose that the output current Iout of the power conversion apparatus **10**A is more than the respective output currents Iout of the power conversion apparatuses **10**B to **10**D. In such a case, in the power conversion apparatus **10**A, a voltage of the detection signal SI**2** is higher than the voltage at the balance terminal Vbi, which makes the difference signal Sdfi smaller, makes the difference value Cdfi smaller, and makes the voltage command value CMV**2** smaller. The output voltage Vout of the

power conversion apparatus **10**A is thereby controlled to be lower, and accordingly, the output current Tout of the power conversion apparatus **10**A is made smaller. In contrast to the above, suppose that the output current Tout of the power conversion apparatus **10**A is less than the respective output currents Tout of the power conversion apparatuses **10**B to **10**D. In such a case, in the power conversion apparatus **10**A, the voltage of the detection signal SI**2** is lower than the voltage at the balance terminal Vbi, which makes the difference signal Sdfi larger, makes the difference value Cdfi larger, and makes the voltage command value CMV**2** larger. The output voltage Vout of the power conversion apparatus **10**A is thereby controlled to be higher, and accordingly, the output current Tout of the power conversion apparatus **10**A is made larger. The same holds true for the power conversion apparatuses **10**B to **10**D. In this way, in the power conversion system **1**, the respective output currents Tout of the power conversion apparatus **10**A to **10**D are controllable to be substantially equal to each other.
(74) Further, in the power conversion system **1**, it is possible to maintain balance between the respective output voltages Vout of the power conversion apparatuses **10**A to **10**D. For example, suppose that the output voltage Vout of the power conversion apparatus **10**A is higher than the respective output voltages Vout of the power conversion apparatuses **10**B to **10**D. In such a case, in the power conversion apparatus **10**A, a voltage of the detection signal SV**2** is higher than the voltage at the balance terminal Vbv, which makes the difference signal Sdfv smaller, makes the difference value Cdfv smaller, and makes the current command value CMI**2** smaller. The output current Tout of the power conversion apparatus **10**A is thereby controlled to be smaller, and accordingly, the output voltage Vout of the power conversion apparatus **10**A is made lower. In contrast to the above, suppose that the output voltage Vout of the power conversion apparatus **10**A is lower than the respective output voltages Vout of the power conversion apparatuses **10**B to **10**D. In such a case, in the power conversion apparatus **10**A, the voltage of the detection signal SV**2** is lower than the voltage at the balance terminal Vbv, which makes the difference signal Sdfv larger, makes the difference value Cdfv larger, and makes the current command value CMI**2** larger. The output current Tout of the power conversion apparatus **10**A is thereby controlled to be larger, and accordingly, the output voltage Vout of the power conversion apparatus **10**A is made higher. The same holds true for the power conversion apparatuses **10**B to **10**D. In this way, in the power conversion system **1**, the respective output voltages Vout of the power conversion apparatus **10**A to **10**D are controllable to be substantially equal to each other.
(75) FIG. **5** illustrates a more specific but non-limiting configuration example of the corrector **101**A and the output circuit **102**A. The corrector **101**A, the photocoupler **13**A, and the output circuit **102**A may configure a signal generation circuit **100**A. Based on the detection signal SI in the secondary-side circuitry supplied from the current sensor **44**, the signal generation circuit **100**A may generate the detection signal SI**2** in the tertiary-side circuitry corresponding to the detection signal SI, while reducing the influence of the characteristic of the photocoupler **13**A.
(76) [Corrector **101**A]
(77) The correction arithmetic circuit **21**A of the corrector **101**A may include an analog-to-digital conversion circuit **121**A, a CTR correction circuit **122**A, a digital-to-analog conversion circuit **123**A, a transmission circuit **124**A, and a reception circuit **125**A.
(78) The analog-to-digital conversion circuit **121**A may be configured to generate a detection value CI which is the digital value of the detection signal SI, by performing, based on the detection signal SI supplied from the current sensor **44**, analog-to-digital conversion at a predetermined sampling frequency.
(79) The CTR correction circuit **122**A may be configured to generate a correction value CCI, based on the detection value CI and a parameter PARA supplied from the reception circuit **125**A. The parameter PARA indicates an estimated value of the current transfer ratio of the photocoupler **13**A. For example, the CTR correction circuit **122**A may generate, based on the parameter PARA, a correction function to which the detection value CI is to be inputted to output the correction value

CCI, and generate the correction value CCI, based on the detection value CI and the correction function. In some embodiments, the CTR correction circuit **122**A may generate, based on the parameter PARA, a lookup table to which the detection value CI is to be inputted to output the correction value CCI, and generate the correction value CCI, based on the detection value CI and the lookup table. The correction value CCI may be, for example, proportional to the detection value CI and vary depending on the detection value CI. For example, the CTR correction circuit **122**A may make the correction value CCI large when the estimated value of the current transfer ratio is small, and may make the correction value CCI small when the estimated value of the current transfer ratio is large.

(80) The digital-to-analog conversion circuit **123**A may be configured to generate the correction signal SCI by performing, based on the correction value CCI generated by the CTR correction circuit **122**A, digital-to-analog conversion at a predetermined sampling frequency. Further, the digital-to-analog conversion circuit **123**A may supply the correction circuit **12**A with the correction signal SCI generated.

(81) The transmission circuit **124**A may be configured to transmit the detection value CI generated by the analog-to-digital conversion circuit **121**A to the correction arithmetic circuit **31**A via the digital isolator **17**.

(82) The reception circuit **125**A may be configured to receive the parameter PARA transmitted from the correction arithmetic circuit **31**A via the digital isolator **17**, and to supply the CTR correction circuit **122**A with the parameter PARA received.

(83) The correction circuit **12**A of the corrector **101**A may include resistors R**1** and R**2**, operational amplifier OPA**1**, and resistors R**3** to R**5**. The resistor R**1** may have a first end to be supplied with the detection signal SI, and a second end coupled to the resistor R**2** and to a positive input terminal of the operational amplifier OPA**1**. The resistor R**2** may have a first end to be supplied with the correction signal SCI, and a second end coupled to the second end of the resistor R**1** and to the positive input terminal of the operational amplifier OPAL The operational amplifier OPA**1** may have the positive input terminal coupled to the second ends of the resistors R**1** and R**2**, a negative input terminal coupled to the resistors R**3** and R**5**, and an output terminal coupled to an anode of the light emitting device of the photocoupler **13**A. In some embodiments, the light emitting device may be a light emitting diode. The resistor R**3** may have a first end coupled to the negative input terminal of the operational amplifier OPA**1** and to the resistor R**5**, and a second end coupled to a reference power supply node of a power supply voltage VGND**2**. This reference power supply node may be a node of a reference power supply of the secondary-side circuitry. The resistor R**4** may have a first end coupled to a cathode of the light emitting device of the photocoupler **13**A and to the resistor R**5**, and a second end coupled to the reference power supply node of the power supply voltage VGND**2**. The resistor R**5** may have a first end coupled to the negative input terminal of the operational amplifier OPA**1** and to the first end of the resistor R**3**, and a second end coupled to the cathode of the light emitting device of the photocoupler **13**A and to the first end of the resistor R**4**.

(84) In the correction circuit **12**A, the detection signal SI and the correction signal SCI may be composited by the resistors R**1** and R**2** to thereby correct the detection signal SI, and the detection signal SI corrected may be supplied to the positive input terminal of the operational amplifier OPAL Further, the correction circuit **12**A may feed a current corresponding to a voltage at the positive input terminal of the operational amplifier OPA**1** through the light emitting device of the photocoupler **13**A.

(85) For example, when the current transfer ratio of the photocoupler **13**A is low, the light reception current at the light receiving device of the photocoupler **13**A can become small. Accordingly, the correction arithmetic circuit **21**A may make the correction signal SCI large to thereby make the voltage at the positive input terminal of the operational amplifier OPA**1** high. This makes it possible to prevent the light reception current at the light receiving device of the photocoupler **13**A from becoming small. Further, when the current transfer ratio of the photocoupler **13**A is high, the

light reception current at the light receiving device of the photocoupler **13**A can become large. Accordingly, the correction arithmetic circuit **21**A may make the correction signal SCI small to thereby make the voltage at the positive input terminal of the operational amplifier OPA**1** low. This makes it possible to prevent the light reception current at the light receiving device of the photocoupler **13**A from becoming large. In such a manner, the corrector **101**A may correct the detection value SI to cause the light reception current at the light receiving device of the photocoupler **13**A to be less susceptible to the current transfer ratio of the photocoupler **13**A.

(86) [Output Circuit **102**A]

(87) The correction circuit **14**A of the output circuit **102**A may include resistors R**6** to R**9** and an operational amplifier OPA**2**. The correction circuit **14**A may couple a collector of the light receiving device of the photocoupler **13**A to a power supply node of a power supply voltage VDD**3**. In some embodiments, the light receiving device may be a phototransistor. The resistor R**6** may have a first end coupled to an emitter of the light receiving device of the photocoupler **13**A and to the resistor R**7**, and a second end be coupled to a reference power supply node of a power supply voltage VGND**3**. This reference power supply node may be a node of a reference power supply of the tertiary-side circuitry, and may be coupled to the reference terminal GNDb illustrated in FIG. **1**. The resistor R**7** may have a first end coupled to the emitter of the light receiving device of the photocoupler **13**A and to the first end of the resistor R**6**, and a second end coupled to the resistors R**8** and R**9** and to a positive input terminal of the operational amplifier OPA**2**. The resistor R**8** may have a first end coupled to the second end of the resistor R**7**, to the resistor R**9**, and to the positive input terminal of the operation amplifier OPA**2**. The resistor R**8** may have a second end coupled to the reference power supply node of the power supply voltage VGND**3**. The resistor R**9** may have a first end to be supplied with the correction signal SCI**2**, and a second end coupled to the second end of the resistor R**7**, to the first end of the resistor R**8**, and to the positive input terminal of the operational amplifier OPA**2**. The positive input terminal of the operational amplifier OPA**2** may be coupled to the second ends of the resistors R**7** and R**9** and to the first end of the resistor R**8**. A negative input terminal of the operational amplifier OPA**2** may be coupled to an output terminal of the operational amplifier OPA**2**. The output terminal of operational amplifier OPA**2** may be coupled to the negative input terminal of the operational amplifier OPA**2**. The operational amplifier OPA**2** may configure a so-called voltage follower circuit, and may generate the detection signal SI**2**. The correction circuit **14**A may supply the detection signal SI**2** to the resistor **15**A.

(88) The correction arithmetic circuit **31**A of the output circuit **102**A may include an analog-to-digital conversion circuit **131**A, a reception circuit **132**A, an output correction circuit **133**A, a digital-to-analog conversion circuit **134**A, a CTR estimation circuit **135**A, and a transmission circuit **136**A. The output correction circuit **133**A and the CTR estimation circuit **135**A may configure a processing circuit **139**A.

(89) The analog-to-digital conversion circuit **131**A may be configured to generate a detection value CI**2** which is a digital value of the detection signal SI**2**, by performing, based on the detection signal SI**2**, analog-to-digital conversion at a predetermined sampling frequency.

(90) The reception circuit **132**A may be configured to receive the detection value CI transmitted from the correction arithmetic circuit **21**A via the digital isolator **17**, and to supply the output correction circuit **133**A with the detection value CI received.

(91) The output correction circuit **133**A may be configured to calculate an expected value of the detection value CI**2**, based on the detection value CI supplied from the reception circuit **132**A, and to generate a correction value CCI**2** to cause the detection value CI**2** supplied from the analog-to-digital conversion circuit **131**A to be equal to the expected value. For example, it is desirable that the detection value CI**2** be equal to the expected value obtained based on the detection value CI. However, the detection value CI**2** can deviate from the expected value even when the corrector **101**A has performed correction processing. To address this, the output correction circuit **133**A may

generate the correction value CCI**2** to cause the detection value CI**2** to be equal to the expected value.
(92) The digital-to-analog conversion circuit **134**A may be configured to generate the correction signal SCI**2** by performing, based on the correction value CCI**2** generated by the output correction circuit **133**A, digital-to-analog conversion at a predetermined sampling frequency. Further, the digital-to-analog conversion circuit **134**A may supply the correction circuit **14**A with the correction signal SCI**2** generated.
(93) The CTR estimation circuit **135**A may be configured to estimate the current transfer ratio of the photocoupler **13**A, based on the correction value CCI**2** generated by the output correction circuit **133**A. Further, the CTR estimation circuit **135**A may supply the parameter PARA indicating the estimated value of the current transfer ratio to the transmission circuit **136**A. Note that although the CTR estimation circuit **135**A may estimate the current transfer ratio based on the correction value CCI**2** in the example embodiment, this is non-limiting. In some embodiments, the CTR estimation circuit **135**A may estimate the current transfer ratio, based on processed data inside the output correction circuit **133**A. In some embodiments, the CTR estimation circuit **135**A may estimate the current transfer ratio, based on detection value CI and the detection value CI**2**.
(94) The transmission circuit **136**A may be configured to transmit the parameter PARA generated by the CTR estimation circuit **135**A to the correction arithmetic circuit **21**A via the digital isolator **17**.
(95) In the correction circuit **14**A, the light reception signal of the light receiving device of the photocoupler **13**A and the correction signal SCI**2** may be composited by the resistors R**7** to R**9**. The output correction circuit **133**A may generate the correction value CCI**2** to cause the detection value CI**2** to be equal to the expected value of the detection value CI**2**. Accordingly, the output circuit **102**A may correct the detection signal SI**2** to cause the detection signal SI**2** to become an expected detection signal SI**2** corresponding to the detection signal SI.
(96) FIG. **6** illustrates a more specific but non-limiting configuration example of the corrector **101**B and the output circuit **102**B. The corrector **101**B, the photocoupler **13**B, and the output circuit **102**B may configure a signal generation circuit **100**B. Based on the detection signal SV in the secondary-side circuitry supplied from the voltage sensor **46**, the signal generation circuit **100**B may generate the detection signal SV**2** in the tertiary-side circuitry corresponding to the detection signal SV, while reducing the influence of the characteristic of the photocoupler **13**B.
(97) [Corrector **101**B]
(98) As with the correction arithmetic circuit **21**A of the corrector **101**A illustrated in FIG. **5**, the correction arithmetic circuit **21**B of the corrector **101**B may include an analog-to-digital conversion circuit **121**B, a CTR correction circuit **122**B, a digital-to-analog conversion circuit **123**B, a transmission circuit **124**B, and a reception circuit **125**B. The analog-to-digital conversion circuit **121**B may be configured to generate a detection value CV which is the digital value of the detection signal SV, by performing, based on the detection signal SV supplied from the voltage sensor **46**, analog-to-digital conversion at a predetermined sampling frequency. The CTR correction circuit **122**B may be configured to generate a correction value CCV, based on the detection value CV and a parameter PARB supplied from the reception circuit **125**B. The parameter PARB indicates an estimated value of a current transfer ratio of the photocoupler **13**B. The digital-to-analog conversion circuit **123**B may be configured to generate the correction signal SCV by performing, based on the correction value CCV generated by the CTR correction circuit **122**B, digital-to-analog conversion at a predetermined sampling frequency. The transmission circuit **124**B may be configured to transmit the detection value CV generated by the analog-to-digital conversion circuit **121**B to the correction arithmetic circuit **31**B via the digital isolator **17**. The reception circuit **125**B may be configured to receive the parameter PARB transmitted from the correction arithmetic circuit **31**B via the digital isolator **17**, and to supply the CTR correction circuit **122**B with the parameter PARB received.

(99) The correction circuit **12**B of the corrector **101**B may have a configuration similar to that of the correction circuit **12**A of the corrector **101**A illustrated in FIG. **5**. The detection signal SV may be supplied to the first end of the resistor R**1**.

(100) [Output Circuit **102**B]

(101) The correction circuit **14**B of the output circuit **102**B may be similar to the correction circuit **14**A of the output circuit **102**A illustrated in FIG. **5**. The correction signal SCV**2** may be supplied to the first end of the resistor R**9**. The operational amplifier OPA**2** may generate the detection signal SV**2**.

(102) As with the correction arithmetic circuit **31**A of the output circuit **102**A illustrated in FIG. **5**, the correction arithmetic circuit **31**B of the output circuit **102**B may include an analog-to-digital conversion circuit **131**B, a reception circuit **132**B, an output correction circuit **133**B, a digital-to-analog conversion circuit **134**B, a CTR estimation circuit **135**B, and a transmission circuit **136**B. The output correction circuit **133**B and the CTR estimation circuit **135**B may configure a processing circuit **139**B. The analog-to-digital conversion circuit **131**B may be configured to generate a detection value CV**2** which is a digital value of the detection signal SV**2**, by performing, based on the detection signal SV**2**, analog-to-digital conversion at a predetermined sampling frequency. The reception circuit **132**B may be configured to receive the detection value CV transmitted from the correction arithmetic circuit **21**B via the digital isolator **17**, and to supply the output correction circuit **133**B with the detection value CV received. The output correction circuit **133**B may be configured to calculate an expected value of the detection value CV**2**, based on the detection value CV supplied from the reception circuit **132**B, and to generate a correction value CCV**2** to cause the detection value CV**2** supplied from the analog-to-digital conversion circuit **131**B to be equal to the expected value. The digital-to-analog conversion circuit **134**B may be configured to generate the correction signal SCV**2** by performing, based on the correction value CCV**2** generated by the output correction circuit **133**B, digital-to-analog conversion at a predetermined sampling frequency. The CTR estimation circuit **135**B may be configured to estimate the current transfer ratio of the photocoupler **13**B, based on the correction value CCV**2**. Further, the CTR estimation circuit **135**B may supply the parameter PARB indicating the estimated value of the current transfer ratio to the transmission circuit **136**B. The transmission circuit **136**B may be configured to transmit the parameter PARB generated by the CTR estimation circuit **135**B to the correction arithmetic circuit **21**B via the digital isolator **17**.

(103) The input power terminals Vip and Vin may each correspond to a specific but non-limiting example of an “input power terminal” in one embodiment of the disclosure. The output power terminals Vop and Von may each correspond to a specific but non-limiting example of an “output power terminal” in one embodiment of the disclosure. The current sensor **44** or the voltage sensor **46** may correspond to a specific but non-limiting example of a “sensor” in one embodiment of the disclosure. The isolated power conversion circuit **11** may correspond to a specific but non-limiting example of a “power conversion circuit” in one embodiment of the disclosure. The switching control circuit **23** may correspond to a specific but non-limiting example of a “control circuit” in one embodiment of the disclosure. The signal generation circuit **100**A or **100**B may correspond to a specific but non-limiting example of a “signal generation circuit” in one embodiment of the disclosure. The detection signal SI or SV may correspond to a specific but non-limiting example of a “first detection signal” in one embodiment of the disclosure. The detection signal SI**2** or SV**2** may correspond to a specific but non-limiting example of a “second detection signal” in one embodiment of the disclosure. The corrector **101**A or **101**B may correspond to a specific but non-limiting example of a “corrector” in one embodiment of the disclosure. The photocoupler **13**A or **13**B may correspond to a specific but non-limiting example of a “photocoupler” in one embodiment of the disclosure. The output circuit **102**A or **102**B may correspond to a specific but non-limiting example of an “output circuit” in one embodiment of the disclosure. The analog-to-digital conversion circuit **121**A or **121**B may correspond to a specific but non-limiting example of a

"first analog-to-digital conversion circuit" in one embodiment of the disclosure. The detection value CI or CV may correspond to a specific but non-limiting example of a "first digital value" in one embodiment of the disclosure. The digital isolator **17** may correspond to a specific but non-limiting example of a "digital isolator" in one embodiment of the disclosure. The analog-to-digital conversion circuit **131**A or **131**B may correspond to a specific but non-limiting example of a "second analog-to-digital conversion circuit" in one embodiment of the disclosure. The detection value CI**2** or CV**2** may correspond to a specific but non-limiting example of a "second digital value" in one embodiment of the disclosure. The processing circuit **139**A or **139**B may correspond to a specific but non-limiting example of a "processing circuit" in one embodiment of the disclosure. The parameter PARA or PARB may correspond to a specific but non-limiting example of a "parameter" in one embodiment of the disclosure. The balance terminals Vbi and Vbv may each correspond to a specific but non-limiting example of a "signal terminal" in one embodiment of the disclosure.

(104) [Operation and Workings]

(105) Operation and workings of the power conversion system **1** according to the present example embodiment will now be described.

(106) [Outline of Overall Operation]

(107) First, an outline of overall operation of the power conversion system **1** will be described with reference to FIGS. **1** and **3**. In each of the four power conversion apparatuses **10**, the isolated power conversion circuit **11** converts direct-current electric power supplied via the input power terminals Vip and Vin, and outputs the converted direct-current electric power via the output power terminals Vop and Von.

(108) Based on the detection signal SI supplied from the current sensor **44**, the signal generation circuit **100**A including the corrector **101**A, the photocoupler **13**A, and the output circuit **102**A may generate the detection signal SI**2** corresponding to the detection signal SI, while reducing the influence of the characteristic of the photocoupler **13**A. The signal generation circuit **100**A may supply, to the first end of the resistor **15**A, the detection signal SI**2** generated. The voltage at the balance terminal Vbi is the average voltage of the detection signals SI**2** of the four power conversion apparatuses **10**.

(109) Based on the detection signal SV supplied from the voltage sensor **46**, the signal generation circuit **100**B including the corrector **101**B, the photocoupler **13**B, and the output circuit **102**B may generate the detection signal SV**2** corresponding to the detection signal SV, while reducing the influence of the characteristic of the photocoupler **13**B. The signal generation circuit **100**B may supply, to the first end of the resistor **15**B, the detection signal SV**2** generated. The voltage at the balance terminal Vbv is the average voltage of the detection signals SV**2** of the four power conversion apparatuses **10**.

(110) The amplifier **16**A may amplify the voltage difference across the resistor **15**A to thereby generate the difference signal Sdfi. The amplifier **16**B may amplify the voltage difference across the resistor **15**B to thereby generate the difference signal Sdfv. Based on the digital value of the difference signal Sdfi, the adjustment arithmetic circuit **32**A may generate the difference value Cdfi that varies depending on the difference signal Sdfi. Based on the digital value of the difference signal Sdfv, the adjustment arithmetic circuit **32**B may generate the difference value Cdfv that varies depending on the difference signal Sdfv. The command value generation circuit **33** may generate the voltage command value CMV of the output voltage Vout of the power conversion apparatus **10** and the current comment value CMI of the output current Iout of the power conversion apparatus **10**. The adder circuit **34**A may generate the voltage command value CMV**2** by adding up the difference value Cdfi and the voltage command value CMV. The adder circuit **34**B may generate the current command value CMI**2** by adding up the difference value Cdfv and the current command value CMI. The error amplifier **22**A may generate the error value Cerv by amplifying the difference between the voltage command value CMV**2** and the digital value of the

detection signal SV. The error amplifier **22**B may generate the error value Ceri by amplifying the difference between the current command value CMI**2** and the digital value of the detection signal SI. The switching control circuit **23** may generate, based on the error values Cerv and Ceri, the gate signal SG**1** and may control the operation of the isolated power conversion circuit **11**, based on the gate signal SG**1**. The isolated drive circuit **18** may generate, based on the gate signal SG**1**, the gate signal SG and may drive the transistors SW**1** and SW**2** of the isolated power conversion circuit **11**, based on the gate signal SG.
(111) [Detailed Operation]
(112) Based on the detection signal SI supplied from the current sensor **44**, the signal generation circuit **100**A including the corrector **101**A, the photocoupler **13**A, and the output circuit **102**A may generate the detection signal SI**2** corresponding to the detection signal SI, while reducing the influence of the characteristic of the photocoupler **13**A. This operation will be described in detail below.
(113) In the secondary-side circuitry, as illustrated in FIG. **5**, the analog-to-digital conversion circuit **121**A may generate the detection value CI which is the digital value of the detection signal SI, by performing, based on the detection signal SI supplied from the current sensor **44**, analog-to-digital conversion at the predetermined sampling frequency. The transmission circuit **124**A may transmit the detection value CI to the tertiary-side circuitry via the digital isolator **17**. The reception circuit **125**A may receive the parameter PARA transmitted from the tertiary-side circuitry via the digital isolator **17**. The CTR correction circuit **122**A may generate the correction value CCI, based on the detection value CI and the parameter PARA indicating the estimated value of the current transfer ratio of the photocoupler **13**A supplied from the reception circuit **125**A. The correction value CCI may be, for example, proportional to the detection value CI and may vary depending on the detection value CI. For example, the CTR correction circuit **122**A may make the correction value CCI large when the estimated value of the current transfer ratio is small, and may make the correction value CCI small when the estimated value of the current transfer ratio is large. Based on the correction value CCI generated by the CTR correction circuit **122**A, the digital-to-analog conversion circuit **123**A may perform digital-to-analog conversion at the predetermined sampling frequency to thereby generate the correction signal SCI.
(114) The correction circuit **12**A may correct the detection signal SI, based on the correction signal SCI generated by the digital-to-analog conversion circuit **123**A, and may drive the photocoupler **13**A, based on the detection signal SI corrected.
(115) The light emitting device of the photocoupler **13**A emits light at a luminance corresponding to the signal supplied from the correction circuit **12**A. The light receiving device receives the light emitted by the light emitting device, and may supply the correction circuit **14**A with the light reception signal corresponding to the amount of the light received from the light emitting device.
(116) In the tertiary-side circuitry, the correction circuit **14**A may generate the detection signal SI**2** corresponding to the light reception signal supplied from the photocoupler **13**A, and may correct the detection signal SI**2**, based on the correction signal SCI**2** supplied from the correction arithmetic circuit **31**A.
(117) The analog-to-digital conversion circuit **131**A may generate the detection value CI**2** which is the digital value of the detection signal SI**2**, by performing, based on the detection signal SI**2** generated by the correction circuit **14**A, analog-to-digital conversion at the predetermined sampling frequency. The reception circuit **132**A may receive the detection value CI transmitted from the secondary-side circuitry via the digital isolator **17**. Based on the detection value CI supplied from the reception circuit **132**A, the output correction circuit **133**A may calculate an expected value of the detection value CI**2** and may generate the correction value CCI**2** to cause the detection value CI**2** supplied from the analog-to-digital conversion circuit **131**A to be equal to the expected value. The digital-to-analog conversion circuit **134**A may generate the correction signal SCI**2** by performing, based on the correction value CCI**2** generated by the output correction circuit **133**A,

digital-to-analog conversion at the predetermined sampling frequency. The correction circuit **14**A may correct the detection signal SI**2**, based on the correction signal SCI**2**.

(118) Based on the correction value CCI**2** generated by the output correction circuit **133**A, the CTR estimation circuit **135**A may estimate the current transfer ratio of the photocoupler **13**A to thereby generate the parameter PARA indicating the estimated value of the current transfer ratio. The transmission circuit **124**A may transmit the parameter PARA to the secondary-side circuitry via the digital isolator **17**.

(119) FIG. **7** illustrates an operation example of the signal generation circuit **100**A. Part (A) of FIG. **7** illustrates an example waveform of the detection signal SI. Part (B) of FIG. **7** illustrates an example waveform of a light emission current If, i.e., a current flowing through the light emitting device of the photocoupler **13**A. Part (C) of FIG. **7** illustrates an example waveform of a light reception current Ic, i.e., a current flowing through the light receiving device of the photocoupler **13**A. Part (D) of FIG. **7** illustrates an example of the current transfer ratio of the photocoupler **13**A. Part (E) of FIG. **7** illustrates an example of the estimated value of the current transfer ratio of the photocoupler **13**A estimated by the CTR estimation circuit **135**A. Part (F) of FIG. **7** illustrates an example waveform of the detection signal SI**2**. FIG. **7** illustrates respective characteristics of the photocoupler **13**A when the current transfer ratio of the photocoupler **13**A is “100”, when the current transfer ratio of the photocoupler **13**A is “200”, and when the current transfer ratio of the photocoupler **13**A is “300”. Note that the current transfer ratio of the photocoupler **13**A may vary also depending on an operating point of the photocoupler **13**A, as indicated in part (D) of FIG. **7**.

(120) In the output circuit **102**A, the CTR estimation circuit **135**A of the correction arithmetic circuit **31**A may estimate the current transfer ratio of the photocoupler **13**A. See part (E) of FIG. **7**.

(121) In the corrector **101**A, the correction arithmetic circuit **21**A may generate the correction signal SCI, based on the detection signal SI illustrated in part (A) of FIG. **7** and the estimated value of the current transfer ratio of the photocoupler **13**A. The detection signal SI may have a sinusoidal shape in the example embodiment, although it is not limited thereto. The correction circuit **12**A may correct the detection signal SI, based on the correction signal SCI generated by the digital-to-analog conversion circuit **123**A, and may drive the photocoupler **13**A, based on the detection signal SI corrected. As a result, as illustrated in part (B) of FIG. **7**, the light emission current If flowing through the light emitting device of the photocoupler **13**A varies depending on the current transfer ratio. For example, the light emission current If may be a large current when the current transfer ratio is “100”, and may be a small current when the current transfer ratio is “300”.

(122) The light emitting device of the photocoupler **13**A emits light at a luminance corresponding to the signal supplied from the correction circuit **12**A. The light receiving device of the photocoupler **13**A receives the light emitted by the light emitting device, and may supply the correction circuit **14**A with the light reception signal corresponding to the amount of the light received from the light emitting device.

(123) As illustrated in part (C) of FIG. **7**, the light reception current Ic flowing through the light receiving device of the photocoupler **13**B is substantially independent of the current transfer ratio. For example, the signal generation circuit **100**A may make the light emission current If large when the current transfer ratio is “100”, and make the light emission current If small when the current transfer ratio is “300”, thereby reducing the influence of the current transfer ratio on the light reception current Ic.

(124) In the output circuit **102**A, the correction circuit **14**A may generate the detection signal SI**2** corresponding to the light reception current Ic, and may correct the detection signal SI**2**, based on the correction signal SCI**2** supplied from the correction arithmetic circuit **31**A. In such a manner, the output circuit **102**A may generate the detection signal SI**2**. See part (F) of FIG. **7**.

(125) In the signal generation circuit **100**A, the corrector **101**A may perform correction processing on the detection signal SI and drive the light emitting device of the photocoupler **13**A, based on the detection signal SI having undergone the correction processing. This makes it possible for the

signal generation circuit **100**A to reduce the influence of the current transfer ratio of the photocoupler **13**A on the detection signal SI**2**, as compared with a comparative example described below.

Comparative Example

(126) A signal generation circuit **100**R according to a comparative example will now be described. The signal generation circuit **100**R differs from the signal generation circuit **100**A in that no correction is performed on the detection signal SI in a stage preceding the photocoupler **13**A.

(127) FIG. **8** illustrates an operation example of the signal generation circuit **100**R. Part (A) of FIG. **8** illustrates an example waveform of the light emission current If, i.e., a current flowing through the light emitting device of the photocoupler **13**A. Part (B) of FIG. **8** illustrates an example waveform of the light reception current Ic, i.e., a current flowing through the light receiving device of the photocoupler **13**A. Part (C) of FIG. **8** illustrates an example of the current transfer ratio of the photocoupler **13**A.

(128) In the signal generation circuit **100**R, no correction is performed on the detection signal SI in the stage preceding the photocoupler **13**A, and the light emission current If corresponding to the detection signal SI flows through the light emitting device of the photocoupler **13**A accordingly. See part (A) of FIG. **8**. The light emission current If is substantially independent of the current transfer ratio.

(129) In contrast, the light reception current Ic of the light receiving device of the photocoupler **13**A varies depending on the current transfer ratio. See part (B) of FIG. **8**. For example, the light reception current Ic is small when the current transfer ratio is “100”, and is large when the current transfer ratio is “300”.

(130) For example, when the current transfer ratio is “300”, the light reception current Ic is large and the correction signal SCI**2** is relatively small in a stage following the photocoupler **13**A accordingly. In other words, the light reception signal of the photocoupler **13**A is dominant as compared with the correction signal SCI**2**. In such a case, the detection signal SI**2** is generated based on the light reception signal of the photocoupler **13**A.

(131) For example, when the current transfer ratio is “100”, the light reception current Ic is small and therefore the correction signal SCI**2** can become dominant in the stage following the photocoupler **13**A. In such a case, the detection signal SI**2** is generated based on the correction signal SCI**2**. The correction signal SCI**2** may be generated through communication between the correction arithmetic circuit **21**A and the correction arithmetic circuit **31**A via the digital isolator **17**. Accordingly, a delay can result from the communication or arithmetic processing, which can cause the detection signal SI**2** to lag in timing behind the detection signal SI.

(132) Thus, according to the signal generation circuit **100**R of the comparative example, a characteristic of the detection signal SI**2** can differ depending on the current transfer ratio. The detection signal SI**2** is a signal corresponding to the output current Iout of the power conversion apparatus **10**. In the signal generation circuit **100**R, detection accuracy for the output current Iout can thus be degraded depending on the current transfer ratio.

(133) In contrast, in the signal generation circuit **100**A according to the present example embodiment, the corrector **101**A may perform the correction processing on the detection signal SI and drive, based on the detection signal SI having undergone the correction processing, the light emitting device of the photocoupler **13**A. This makes it possible to maintain the light reception current Ic not to become small irrespective of the current transfer ratio, as illustrated in FIG. **7**. Accordingly, for example, at the correction circuit **14**A disposed in the stage following the photocoupler **13**A, it is possible to make the light reception signal of the photocoupler **13**A sufficiently large as compared with the correction signal SCI**2**. This causes the detection signal SI**2** to be a signal corresponding to the detection signal SI irrespective of the current transfer ratio of the photocoupler **13**A. As a result, the signal generation circuit **100**A makes it possible to increase the detection accuracy for the output current Iout.

(134) The foregoing description has dealt with the signal generation circuit **100**A as an example; however, a similar description applies to the signal generation circuit **100**B. Accordingly, the signal generation circuit **100**B makes it possible to increase detection accuracy for the output voltage Vout.
(135) According to the example embodiment described above, the power conversion apparatus **10** includes the input power terminals Vip and Vin, the output power terminals Vop and Von, the isolated power conversion circuit **11**, the signal generation circuit **100**A, and the switching control circuit **23**. The isolated power conversion circuit **11** converts electric power supplied via the input power terminals Vip and Vin, and outputs converted electric power via the output power terminals Vop and Von. The isolated power conversion circuit **11** includes the current sensor **44** that generates the detection signal SI corresponding to the output current. The signal generation circuit **100**A generates the detection signal SI**2** corresponding to the detection signal SI. The switching control circuit **23** controls the operation of the isolated power conversion circuit **11**. The signal generation circuit **100**A includes the corrector **101**A, the photocoupler **13**A, and the output circuit **102**A. The corrector **101**A performs correction processing on the detection signal SI. The photocoupler **13**A includes the light emitting device and the light receiving device. The light emitting device emits light at a luminance corresponding to the detection signal SI having undergone the correction processing. The light receiving device receives the light emitted by the light emitting device and generates the light reception signal corresponding to the amount of the light received from the light emitting device. The output circuit **102**A outputs the detection signal SI**2** corresponding to the light reception signal. The corrector **101**A performs the correction processing corresponding to the current transfer ratio of the photocoupler **13**A. As a result, it is possible for the power conversion apparatus **10** to maintain the light reception current Ic not to become small irrespective of the current transfer ratio of the photocoupler **13**A, as illustrated in FIG. **7**. Accordingly, it is possible to increase the detection accuracy for the output current Iout.
(136) In some embodiments, the signal generation circuit **100**A may include the analog-to-digital conversion circuit **121**A, the digital isolator **17**, the analog-to-digital conversion circuit **131**A, and the processing circuit **139**A. The analog-to-digital conversion circuit **121**A may generate the detection value CI by performing analog-to-digital conversion on the detection signal SI. The analog-to-digital conversion circuit **131**A may generate the detection value CI**2** by performing analog-to-digital conversion on the detection signal SI**2**. Based on the detection value CI supplied via the digital isolator **17** and the detection value CI**2**, the processing circuit **139**A may estimate the current transfer ratio and may generate the parameter PARA corresponding to the estimated current transfer ratio. The corrector **101**A may perform, based on the parameter PARA, the correction processing corresponding to the current transfer ratio. It is thus possible for the signal generation circuit **100**A to perform the correction processing on the detection signal SI using the estimated current transfer ratio. Accordingly, for example, even when the current transfer ratio changes with temperature or with time, the corrector **101**A is able to perform the correction processing effectively on the detection signal SI, in accordance with the change. This makes it possible for the power conversion apparatus **10** to increase the detection accuracy for the output current Iout.
(137) In some embodiments, the power conversion apparatus **10** may further include the balance terminal Vbi led to a terminal, the terminal allowing for output of the detection signal SI**2** of the signal generation circuit **100**A. In addition, the switching control circuit **23** may control the operation of the isolated power conversion circuit **11**, based on the voltage at the balance terminal Vbi and the voltage of the detection signal SI**2**. As a result, for example, when a plurality of power conversion apparatuses **10** is provided, it is possible to make the respective output currents Iout of the power conversion apparatuses **10** substantially equal to each other.
(138) In some embodiments, the plurality of power conversion apparatuses **10** may include the power conversion apparatus **10**A and the power conversion apparatus **10**B, for example. The power conversion apparatus **10**A may include the output power terminal Vop coupled to the power node

N**1** led to the power terminal T**21**, and the output power terminal Von coupled to the power node N**2**. The power conversion apparatus **10**B may include the output power terminal Vop coupled to the power node N**1**, and the output power terminal Von coupled to the power node N**2**. In other words, the power conversion apparatuses **10**A and **10**B may be coupled in parallel to each other. In such a case, it is possible to make the output current Tout of the power conversion apparatus **10**A and the output current Tout of the power conversion apparatus **10**B substantially equal to each other.

(139) In the power conversion apparatus **10** according to the example embodiment, the isolated power conversion circuit **11** includes the voltage sensor **46** that generates the detection signal SV corresponding to the output voltage. In addition, the power conversion apparatus **10** includes the signal generation circuit **100**B that generates the detection signal SV**2** corresponding to the detection signal SV. The signal generation circuit **100**B includes the corrector **101**B, the photocoupler **13**B, and the output circuit **102**B. The corrector **101**B performs correction processing on the detection signal SV. The photocoupler **13**B includes the light emitting device and the light receiving device. The light emitting device emits light at a luminance corresponding to the detection signal SV having undergone the correction processing. The light receiving device receives the light emitted by the light emitting device and generates the light reception signal corresponding to the amount of the light received from the light emitting device. The output circuit **102**B outputs the detection signal SV**2** corresponding to the light reception signal. The corrector **101**B performs the correction processing corresponding to the current transfer ratio of the photocoupler **13**B. As a result, it is possible for the power conversion apparatus **10** to maintain the light reception current Ic not to become small irrespective of the current transfer ratio of the photocoupler **13**B. Accordingly, it is possible to increase the detection accuracy for the output voltage Vout.

(140) In some embodiments, the signal generation circuit **100**B may include the analog-to-digital conversion circuit **121**B, the digital isolator **17**, the analog-to-digital conversion circuit **131**B, and the processing circuit **139**B. The analog-to-digital conversion circuit **121**B may generate the detection value CV by performing analog-to-digital conversion on the detection signal SV. The analog-to-digital conversion circuit **131**B may generate the detection value CV**2** by performing analog-to-digital conversion on the detection signal SV**2**. Based on the detection value CV supplied via the digital isolator **17** and the detection value CV**2**, the processing circuit **139**B may estimate the current transfer ratio and may generate the parameter PARB corresponding to the estimated current transfer ratio. The corrector **101**B may perform, based on the parameter PARB, the correction processing corresponding to the current transfer ratio. It is thus possible for the signal generation circuit **100**B to perform the correction processing on the detection signal SV using the estimated current transfer ratio. Accordingly, for example, even when the current transfer ratio changes with temperature or with time, the corrector **101**B is able to perform the correction processing effectively on the detection signal SV, in accordance with the change. This makes it possible for the power conversion apparatus **10** to increase the detection accuracy for the output voltage Vout.

(141) In some embodiments, the power conversion apparatus **10** may further include the balance terminal Vbv led to a terminal, the terminal allowing for output of the detection signal SV**2** of the signal generation circuit **100**B. In addition, the switching control circuit **23** may control the operation of the isolated power conversion circuit **11**, based on the voltage at the balance terminal Vbv and the voltage of the detection signal SV**2**. As a result, for example, when a plurality of power conversion apparatuses **10** is provided, it is possible to make the respective output voltages Vout of the power conversion apparatuses **10** substantially equal to each other.

(142) In some embodiments, the plurality of power conversion apparatuses **10** may include the power conversion apparatus **10**A and the power conversion apparatus **10**C, for example. The power conversion apparatus **10**A may include the output power terminal Vop coupled to the power node

N**1** led to the power terminal T**21**, and the output power terminal Von coupled to the power node N**2**. The power conversion apparatus **10**C may include the output power terminal Vop coupled to the power node N**2**, and the output power terminal Von coupled to the power node N**3** led to the power terminal T**22**. In other words, the power conversion apparatuses **10**A and **10**C may be coupled in series to each other. In such a case, it is possible to make the output voltage Vout of the power conversion apparatus **10**A and the output voltage Vout of the power conversion apparatus **10**C substantially equal to each other.

Example Effects

(143) The foregoing example embodiment includes the input power terminal, the output power terminal, the isolated power conversion circuit, the signal generation circuit, and the switching control circuit. The isolated power conversion circuit converts electric power supplied via the input power terminal, and outputs the converted electric power via the output power terminal. The isolated power conversion circuit includes the current sensor that generates the detection signal SI corresponding to the output current. The signal generation circuit generates the detection signal SI**2** corresponding to the detection signal SI. The switching control circuit controls the operation of the isolated power conversion circuit. The signal generation circuit includes the corrector, the photocoupler, and the output circuit. The corrector performs correction processing on the detection signal SI. The photocoupler includes the light emitting device and the light receiving device. The light emitting device emits light at a luminance corresponding to the detection signal SI having undergone the correction processing. The light receiving device receives the light emitted by the light emitting device and generates the light reception signal corresponding to the amount of the light received from the light emitting device. The output circuit outputs the detection signal SI**2** corresponding to the light reception signal. The corrector performs the correction processing corresponding to the current transfer ratio of the photocoupler. This helps to increase the detection accuracy for the output current.

(144) In some embodiments, the signal generation circuit may include the analog-to-digital conversion circuit **121**A, the digital isolator, the analog-to-digital conversion circuit **131**A, and the processing circuit. The analog-to-digital conversion circuit **121**A may generate the detection value CI by performing analog-to-digital conversion on the detection signal SI. The analog-to-digital conversion circuit **131**A may generate the detection value CI**2** by performing analog-to-digital conversion on the detection signal SI**2**. The processing circuit may, based on the detection value CI supplied via the digital isolator and the detection value CI**2**, estimate the current transfer ratio and may generate the parameter corresponding to the estimated current transfer ratio. The corrector may perform, based on the parameter, the correction processing corresponding to the current transfer ratio. This helps to increase the detection accuracy for the output current.

(145) In some embodiments, the isolated power conversion circuit includes the voltage sensor that generates the detection signal SV corresponding to the output voltage. In addition, the signal generation circuit is provided that generates the detection signal SV**2** corresponding to the detection signal SV. The signal generation circuit includes the corrector, the photocoupler, and the output circuit. The corrector performs correction processing on the detection signal SV. The photocoupler includes the light emitting device and the light receiving device. The light emitting device emits light at a luminance corresponding to the detection signal SV having undergone the correction processing. The light receiving device receives the light emitted by the light emitting device and generates the light reception signal corresponding to the amount of the light received from the light emitting device. The output circuit outputs the detection signal SV**2** corresponding to the light reception signal. The corrector performs the correction processing corresponding to the current transfer ratio of the photocoupler. This helps to increase the detection accuracy for the output voltage.

(146) In some embodiments, the signal generation circuit may include the analog-to-digital conversion circuit **121**B, the digital isolator, the analog-to-digital conversion circuit **131**B, and the

processing circuit. The analog-to-digital conversion circuit **121**B may generate the detection value CV by performing analog-to-digital conversion on the detection signal SV. The analog-to-digital conversion circuit **131**B may generate the detection value CV**2** by performing analog-to-digital conversion on the detection signal SV**2**. The processing circuit may, based on the detection value CV supplied via the digital isolator **17** and the detection value CV**2**, estimate the current transfer ratio and may generate the parameter corresponding to the estimated current transfer ratio. The corrector may perform, based on the parameter, the correction processing corresponding to the current transfer ratio. This helps to increase the detection accuracy for the output voltage.

Modification Example 1

(147) In the foregoing example embodiment, the power conversion apparatus **10** may include the isolated power conversion circuit **11**; however, this is non-limiting. In some embodiments, the power conversion apparatus may include a non-isolated power conversion circuit. A power conversion system **2** according to the present modification example will be described in detail below.

(148) FIG. **9** illustrates a configuration example of the power conversion system **2**. The power conversion system **2** may include a plurality of power conversion apparatuses **50** and a plurality of power conversion apparatuses **60**. In this modification example, the power conversion system **2** may include four power conversion apparatuses **50**A, **50**B, **50**C, and **50**D, and four power conversion apparatuses **60**A, **60**B, **60**C, and **60**D, although they are not limited thereto. The four power conversion apparatuses **50** and the four power conversion apparatuses **60** may be provided in correspondence with each other.

(149) The four power conversion apparatuses **50** may each be an isolated DC-to-DC conversion circuit. The power conversion apparatuses **50** each include the input power terminals Vip and Vin and the output power terminals Vop and Von. The respective input power terminals Vip of the power conversion apparatuses **50**A to **50**D may be coupled to each other, and may also be coupled to the power terminal T**11**. The respective input power terminals Vin of the power conversion apparatuses **50**A to **50**D may be coupled to each other, and may also be coupled to the power terminal T**12**. The output power terminal Vop of each of the power conversion apparatuses **50** may be coupled to the input power terminal Vip of corresponding one of the power conversion apparatuses **60**. The output power terminal Von of each of the power conversion apparatuses **50** may be coupled to the input power terminal Vin of corresponding one of the power conversion apparatuses **60**. The power conversion apparatuses **50** may each be an isolated circuit, and may therefore include primary-side circuitry, a transformer, and secondary-side circuitry, as with the isolated power conversion circuit **11** illustrated in FIG. **4**, for example.

(150) The four power conversion apparatuses **60** may each be a non-isolated DC-to-DC conversion circuit. The power conversion apparatuses **60** may each include the input power terminals Vip and Vin, the output power terminals Vop and Von, the balance terminals Vbi and Vbv, and the reference terminal GNDb.

(151) The input power terminal Vip of each of the power conversion apparatuses **60** may be coupled to the output power terminal Vop of corresponding one of the power conversion apparatuses **50**. The input power terminal Vin of each of the power conversion apparatuses **60** may be coupled to the output power terminal Von of corresponding one of the power conversion apparatuses **50**.

(152) The output power terminals Vop and Von of the power conversion apparatuses **60** may be coupled in a manner similar to that in the power conversion system **1** according to the foregoing example embodiment illustrated in FIG. **1**. For example, the respective output power terminals Vop of the power conversion apparatuses **60**A and **60**B may be coupled to each other, and may also be coupled to the power terminal T**21**. The respective output power terminals Von of the power conversion apparatuses **60**A and **60**B may be coupled to each other, and may also be coupled to the output power terminal Vop of each of the power conversion apparatuses **60**C and **60**D. The

respective output power terminals Vop of the power conversion apparatuses **60**C and **60**D may be coupled to each other, and may also be coupled to the output power terminal Von of each of the power conversion apparatuses **60**A and **60**B. The respective output power terminals Von of the power conversion apparatuses **60**C and **60**D may be coupled to each other, and may also be coupled to the power terminal T**22**. In the power conversion system **2**, as in the power conversion system **1** according to the foregoing example embodiment (see FIG. **2**), the power conversion apparatuses **60**A and **60**B may be coupled in parallel, and the power conversion apparatuses **60**C and **60**D may be coupled in parallel. Further, the power conversion apparatuses **60**A and **60**B and the power conversion apparatuses **60**C and **60**D may be coupled in series.
(153) The respective balance terminals Vbi of the power conversion apparatuses **60**A to **60**D are couped to each other. The respective balance terminals Vbv of the power conversion apparatuses **60**A to **60**D are couped to each other. The respective reference terminals GNDb of the power conversion apparatuses **60**A to **60**D may be couped to each other.
(154) FIG. **10** illustrates a configuration example of the power conversion apparatus **60**. The power conversion apparatus **60** may include a non-isolated power conversion circuit **61** and a drive circuit **68**.
(155) FIG. **11** illustrates a configuration example of the non-isolated power conversion circuit **61**. The non-isolated power conversion circuit **61** may include a capacitor C**3**, a transistor SW**3**, a rectifier circuit **73**, a current sensor **74**, a smoothing circuit **75**, and a voltage sensor **76**.
(156) The capacitor C**3** may have a first end coupled to a voltage line L**31**, and a second end coupled to a reference voltage line L**32**. The voltage line L**31** may be led to the input power terminal Vip and the output power terminal Vop. The reference voltage line L**32** may be led to the input power terminal Vin and the output power terminal Von. The transistor SW**3** may be an N-type field-effect transistor in this modification example, although it is not limited thereto. The transistor SW**3** may be provided on the voltage line L**31**. The transistor SW**3** may have a gate to be supplied with the gate signal SG, a drain coupled to the first end of the capacitor C**3**, and a source coupled to the rectifier circuit **73**. The rectifier circuit **73** may include a diode D**3**. The diode D**3** may have an anode coupled to the reference voltage line L**32**, and a cathode coupled to the source of the transistor SW**3** on the voltage line L**31**. The current sensor **74** may be configured to detect the output current Tout of the power conversion apparatus The current sensor **74** may be provided on the reference voltage line L**32**. The current sensor **74** may have a first end coupled to the anode of the diode D**3** and to the second end of the capacitor C**3**, and a second end coupled to the smoothing circuit **75**. The smoothing circuit **75** may include an inductor L**2** and a capacitor C**4**. The inductor L**2** may be provided on the voltage line L**31**. The inductor L**2** may have a first end coupled to the cathode of the diode D**3** and to the source of the transistor SW**3**, and a second end coupled to the capacitor C**4**. The capacitor C**4** may have a first end coupled to the second end of the inductor L**2** on the voltage line L**31**, and a second end coupled to the second end of the current sensor **74** on the reference voltage line L**32**. The voltage sensor **76** may be configured to detect the output voltage Vout of the power conversion apparatus **60**. The voltage sensor **76** may have a first end coupled to the second end of the inductor L**2** on the voltage line L**31**, and a second end coupled to the second end of the current sensor **74** on the reference voltage line L**32**.
(157) As illustrated in FIG. **9**, the four power conversion apparatuses **50**A to **50**D may be supplied with direct-current electric power from the direct-current power supply PDC. Accordingly, operating voltages of the primary-side circuitry of the power conversion apparatuses **50**A to **50**D may be equal to each other.
(158) In FIG. **10**, the non-isolated power conversion circuit **61**, the correction circuits **12**A and **12**B, the MCU **20**, and the drive circuit **68** may configure the secondary-side circuitry. The correction circuits **14**A and **14**B, the resistors **15**A and **15**B, the amplifiers **16**A and **16**B, and the MCU **30** may configure tertiary-side circuitry.
(159) As in the foregoing power conversion system **1** (see FIG. **2**), the power conversion

apparatuses **60**A and **60**B and the power conversion apparatuses **60**C and **60**D may be coupled in series. Accordingly, operating voltages, based on the voltage at the power terminal T**22**, of the secondary-side circuitry of the power conversion apparatuses **60**A and **60**B may be higher than operating voltages, based on the voltage at the power terminal T**22**, of the secondary-side circuitry of the power conversion apparatuses **60**C and **60**D.

(160) As illustrated in FIG. **9**, the respective reference terminals GNDb of the four power conversion apparatuses **60** may be coupled to each other. Accordingly, reference voltages of the tertiary-side circuitry of the four power conversion apparatuses **60** may be the same, and therefore operating voltages of the tertiary-side circuitry of the four power conversion apparatuses **60** may be equal to each other.

(161) The drive circuit **68** illustrated in FIG. **10** may be configured to generate the gate signal SG, based on the gate signal SG**1**, and to drive the transistor SW**3** (see FIG. **11**) of the non-isolated power conversion circuit **61**, based on the gate signal SG. The MCU **20** generating the gate signal SG**1** and the transistor SW**3** to be supplied with the gate signal SG may both be elements of the secondary-side circuitry. Accordingly, the drive circuit **68** may drive the transistor SW**3**, based on the gate signal SG**1**, without electrically isolating the MCU **20** and the transistor SW**3** from each other.

(162) With such a configuration, in the power conversion apparatus **60**, as in the power conversion apparatus **10** according to the foregoing example embodiment, negative feedback control may be performed to cause the output voltage Vout detected by the voltage sensor **76** to be equal to a voltage indicated by the voltage command value CMV**2**, and negative feedback control may be performed to cause the output current Iout detected by the current sensor **74** to be equal to a current indicated by the current command value CMI**2**.

(163) Further, as with the power conversion system **1** according to the foregoing example embodiment, the power conversion system **2** helps to maintain balance between the respective output currents Tout of the power conversion apparatuses **60**A to **60**D, and to maintain balance between the respective output voltages Vout of the power conversion apparatuses **60**A to **60**D.

Modification Example 2

(164) In the foregoing example embodiment, the estimated value of the current transfer ratio of the photocoupler **13**A may be used as the parameter PARA; however, this is non-limiting. In some embodiments, any of various parameters corresponding to the estimated value of the current transfer ratio of the photocoupler **13**A may be used as the parameter PARA. For example, when the CTR correction circuit **122**A generates the correction value CCI, based on a correction function to which the detection value CI is to be inputted to output the correction value CCI, a parameter indicating the correction function may be used as the parameter PARA. Further, for example, when the CTR correction circuit **122**A generates the correction value CCI, based on a lookup table to which the detection value CI is to be inputted to output the correction value CCI, a parameter indicating the lookup table may be used as the parameter PARA. Similarly, although the estimated value of the current transfer ratio of the photocoupler **13**B may be used as the parameter PARB, this is non-limiting. In some embodiments, any of various parameters corresponding to the estimated value of the current transfer ratio of the photocoupler **13**B may be used as the parameter PARB.

Modification Example 3

(165) In the foregoing example embodiment, as illustrated in FIGS. **1** and **2**, the power conversion apparatuses **10**A and **10**B may be coupled in parallel, the power conversion apparatuses **10**C and **10**D may be coupled in parallel, and the power conversion apparatuses **10**A and **10**B and the power conversion apparatuses **10**C and **10**D may be coupled in series in the power conversion system **1**; however, this is non-limiting. In some embodiments, as in a power conversion system **1**A illustrated in FIGS. **12** and **13**, the power conversion apparatuses **10**A and may be coupled in series, the power conversion apparatuses **10**B and **10**D may be coupled in series, and the power

conversion apparatuses **10**A and **10**C and the power conversion apparatuses **10**B and **10**D may be coupled in parallel. Note that the present modification example is applied to the power conversion system **1** illustrated in FIGS. **1** and **2**. In some embodiments, the present modification example may be applied to the power conversion system **2** illustrated in FIG. **9**.

Other Modification Examples

(166) Any two or more of the foregoing modification examples may be employed in combination. Further, the disclosure encompasses any possible combination of some or all of the various embodiments described herein and incorporated herein.

(167) The disclosure has been described hereinabove with reference to the example embodiment and the modification examples. However, the disclosure is not limited thereto, and various modifications may be made.

(168) For example, in the foregoing example embodiment and modification examples, the disclosure may be applied to the power conversion apparatus including the power conversion circuit having the circuit configuration illustrated in, for example, FIG. **4** or **11**; however, this is non-limiting. In some embodiments, the disclosure may be applied to any of power conversion apparatuses having various circuit configurations to which the disclosure is applicable.

(169) For example, in the foregoing example embodiment, the signal generation circuits **100**A and **100**B may both be provided; however, this is non-limiting. In some embodiments, either the signal generation circuit **100**A or the signal generation circuit **100**B may only be provided.

(170) For example, in the foregoing example embodiment and modification examples, as illustrated in FIGS. **1** and **2**, the four power conversion apparatuses **10** may be provided; however, this is non-limiting. In some embodiments, one or more power conversion apparatuses **10** may be provided. In some embodiments where two or more power conversion apparatuses **10** are provided, the power conversion apparatuses **10** may be coupled in series or in parallel to each other. Further, as illustrated in FIG. **2**, pieces of equipment each including two or more parallel-coupled power conversion apparatuses **10** may be coupled in series to each other. Similarly, in the foregoing example embodiment, although the four power conversion apparatuses **60** may be provided as illustrated in FIG. **9**, this is non-limiting. In some embodiments, one or more power conversion apparatuses **60** may be provided.

(171) It is possible to achieve at least the following configurations from the foregoing example embodiment and modification examples of the disclosure.

(172) (1)

(173) A power conversion apparatus including: an input power terminal; an output power terminal; a power conversion circuit configured to convert electric power supplied via the input power terminal and to output converted electric power via the output power terminal, the power conversion circuit including a sensor configured to generate a first detection signal corresponding to an output voltage or an output current; a signal generation circuit configured to generate a second detection signal corresponding to the first detection signal; and a control circuit configured to control operation of the power conversion circuit, in which the signal generation circuit includes a corrector configured to perform correction processing on the first detection signal, a photocoupler including a light emitting device and a light receiving device, the light emitting device being configured to emit light at a luminance corresponding to the first detection signal having undergone the correction processing, the light receiving device being configured to receive the light emitted by the light emitting device and to generate a light reception signal corresponding to an amount of the light received from the light emitting device, and an output circuit configured to output the second detection signal corresponding to the light reception signal, and the corrector is configured to perform the correction processing corresponding to a current transfer ratio of the photocoupler.

(174) (2)

(175) The power conversion apparatus according to (1), in which the signal generation circuit includes a first analog-to-digital conversion circuit configured to generate a first digital value by

performing analog-to-digital conversion on the first detection signal, a digital isolator, a second analog-to-digital conversion circuit configured to generate a second digital value by performing analog-to-digital conversion on the second detection signal, and a processing circuit configured to estimate the current transfer ratio, based on the first digital value supplied via the digital isolator and the second digital value, and to generate a parameter corresponding to the current transfer ratio estimated, and the corrector is configured to perform, based on the parameter, the correction processing corresponding to the current transfer ratio.

(176) (3)

(177) The power conversion apparatus according to (2), in which the processing circuit is configured to correct the second detection signal, based on the first digital value supplied via the digital isolator and the second digital value.

(178) (4)

(179) The power conversion apparatus according to any one of (1) to (3), further including a signal terminal led to a terminal, the terminal allowing for output of the second detection signal of the signal generation circuit, in which the control circuit is configured to control the operation of the power conversion circuit, based on a voltage at the signal terminal and a voltage of the second detection signal.

(180) (5)

(181) A power conversion system including a plurality of power conversion apparatuses, each of the power conversion apparatuses including: an input power terminal; an output power terminal; a power conversion circuit configured to convert electric power supplied via the input power terminal and to output converted electric power via the output power terminal, the power conversion circuit including a sensor configured to generate a first detection signal corresponding to an output voltage or an output current; a signal generation circuit configured to generate a second detection signal corresponding to the first detection signal; a signal terminal led to a terminal, the terminal allowing for output of the second detection signal of the signal generation circuit; and a control circuit configured to control operation of the power conversion circuit, based on a voltage at the signal terminal and a voltage of the second detection signal, in which the respective signal terminals of the power conversion apparatuses are coupled to each other, the signal generation circuit includes a corrector configured to perform correction processing on the first detection signal, a photocoupler including a light emitting device and a light receiving device, the light emitting device being configured to emit light at a luminance corresponding to the first detection signal having undergone the correction processing, the light receiving device being configured to receive the light emitted by the light emitting device and to generate a light reception signal corresponding to an amount of the light received from the light emitting device, and an output circuit configured to output the second detection signal corresponding to the light reception signal, and the corrector is configured to perform the correction processing corresponding to a current transfer ratio of the photocoupler.

(182) (6)

(183) The power conversion system according to (5), in which the output power terminal of each of the power conversion apparatuses includes a power terminal and a reference power terminal, and the power conversion apparatuses include a first power conversion apparatus having the power terminal coupled to a first power node and the reference power terminal coupled to a second power node, and a second power conversion apparatus having the power terminal coupled to the first power node and the reference power terminal coupled to the second power node.

(184) (7)

(185) The power conversion system according to (5) or (6), in which the output power terminal of each of the power conversion apparatuses includes a power terminal and a reference power terminal, and the power conversion apparatuses include a first power conversion apparatus having the power terminal coupled to a first power node and the reference power terminal coupled to a

second power node, and a third power conversion apparatus having the power terminal coupled to the second power node and the reference power terminal coupled to a third power node.
(186) The power conversion apparatus and the power conversion system according to at least one embodiment of the disclosure each make it possible to increase detection accuracy.
(187) Although the technology has been described hereinabove in terms of the example embodiment and modification examples, the technology is not limited thereto. It should be appreciated that variations may be made in the described example embodiment and modification examples by those skilled in the art without departing from the scope of the disclosure as defined by the following claims. The limitations in the claims are to be interpreted broadly based on the language employed in the claims and not limited to examples described in this specification or during the prosecution of the application, and the examples are to be construed as non-exclusive. The use of the terms first, second, etc. do not denote any order or importance, but rather the terms first, second, etc. are used to distinguish one element from another. The term “substantially” and its variants are defined as being largely but not necessarily wholly what is specified as understood by one of ordinary skill in the art. The term “disposed on/provided on/formed on” and its variants as used herein refer to elements disposed directly in contact with each other or indirectly by having intervening structures therebetween. Moreover, no element or component in this disclosure is intended to be dedicated to the public regardless of whether the element or component is explicitly recited in the following claims.

Claims

1. A power conversion apparatus comprising: an input power terminal; an output power terminal; a power conversion circuit configured to convert electric power supplied via the input power terminal and to output converted electric power via the output power terminal, the power conversion circuit including a sensor configured to generate a first detection signal corresponding to an output voltage or an output current; a signal generation circuit configured to generate a second detection signal corresponding to the first detection signal; and a control circuit configured to control operation of the power conversion circuit, wherein the signal generation circuit includes a corrector configured to perform correction processing on the first detection signal, a photocoupler including a light emitting device and a light receiving device, the light emitting device being configured to emit light at a luminance corresponding to the first detection signal having undergone the correction processing, the light receiving device being configured to receive the light emitted by the light emitting device and to generate a light reception signal corresponding to an amount of the light received from the light emitting device, and an output circuit configured to output the second detection signal corresponding to the light reception signal, the corrector is configured to perform the correction processing corresponding to a current transfer ratio of the photocoupler, the signal generation circuit includes a first analog-to-digital conversion circuit configured to generate a first digital value by performing analog-to-digital conversion on the first detection signal, a digital isolator, a second analog-to-digital conversion circuit configured to generate a second digital value by performing analog-to-digital conversion on the second detection signal, and a processing circuit configured to estimate the current transfer ratio, based on the first digital value supplied via the digital isolator and the second digital value, and to generate a parameter corresponding to the current transfer ratio estimated, and the corrector is configured to perform, based on the parameter, the correction processing corresponding to the current transfer ratio.
2. The power conversion apparatus according to claim 1, wherein the processing circuit is configured to correct the second detection signal, based on the first digital value supplied via the digital isolator and the second digital value.
3. The power conversion apparatus according to claim 1, further comprising a signal terminal led to a terminal, the terminal allowing for output of the second detection signal of the signal generation

circuit, wherein the control circuit is configured to control the operation of the power conversion circuit, based on a voltage at the signal terminal and a voltage of the second detection signal.

4. The power conversion apparatus according to claim 2, further comprising a signal terminal led to a terminal, the terminal allowing for output of the second detection signal of the signal generation circuit, wherein the control circuit is configured to control the operation of the power conversion circuit, based on a voltage at the signal terminal and a voltage of the second detection signal.

5. A power conversion system comprising a plurality of power conversion apparatuses, each of the power conversion apparatuses including: an input power terminal; an output power terminal; a power conversion circuit configured to convert electric power supplied via the input power terminal and to output converted electric power via the output power terminal, the power conversion circuit including a sensor configured to generate a first detection signal corresponding to an output voltage or an output current; a signal generation circuit configured to generate a second detection signal corresponding to the first detection signal; a signal terminal led to a terminal, the terminal allowing for output of the second detection signal of the signal generation circuit; and a control circuit configured to control operation of the power conversion circuit, based on a voltage at the signal terminal and a voltage of the second detection signal, wherein the respective signal terminals of the power conversion apparatuses are coupled to each other, the signal generation circuit includes a corrector configured to perform correction processing on the first detection signal, a photocoupler including a light emitting device and a light receiving device, the light emitting device being configured to emit light at a luminance corresponding to the first detection signal having undergone the correction processing, the light receiving device being configured to receive the light emitted by the light emitting device and to generate a light reception signal corresponding to an amount of the light received from the light emitting device, and an output circuit configured to output the second detection signal corresponding to the light reception signal, the corrector is configured to perform the correction processing corresponding to a current transfer ratio of the photocoupler, the signal generation circuit includes a first analog-to-digital conversion circuit configured to generate a first digital value by performing analog-to-digital conversion on the first detection signal, a digital isolator, a second analog-to-digital conversion circuit configured to generate a second digital value by performing analog-to-digital conversion on the second detection signal, and a processing circuit configured to estimate the current transfer ratio, based on the first digital value supplied via the digital isolator and the second digital value, and to generate a parameter corresponding to the current transfer ratio estimated, and the corrector is configured to perform, based on the parameter, the correction processing corresponding to the current transfer ratio.

6. The power conversion system according to claim 5, wherein the output power terminal of each of the power conversion apparatuses includes a power terminal and a reference power terminal, and the power conversion apparatuses include a first power conversion apparatus having the power terminal coupled to a first power node and the reference power terminal coupled to a second power node, and a second power conversion apparatus having the power terminal coupled to the first power node and the reference power terminal coupled to the second power node.

7. The power conversion system according to claim 5, wherein the output power terminal of each of the power conversion apparatuses includes a power terminal and a reference power terminal, and the power conversion apparatuses include a first power conversion apparatus having the power terminal coupled to a first power node and the reference power terminal coupled to a second power node, and a third power conversion apparatus having the power terminal coupled to the second power node and the reference power terminal coupled to a third power node.

8. The power conversion system according to claim 6, wherein the power conversion apparatuses further include a third power conversion apparatus having the power terminal coupled to the second power node and the reference power terminal coupled to a third power node.